

IDENTIFICATION

PRODUCT CODE: MAINDEC-8E-D0FC-D
PRODUCT NAME: RANDOM ISE TEST
DATE CREATED: JUNE 11, 1971
MAINTAINER: DIAGNOSTIC GROUP
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1. ABSTRACT

THIS PROGRAM IS WRITTEN TO TEST THE ISZ INSTRUCTION OF THE PDP-8E. AN ISZ INSTRUCTION IS PLACED IN A FROM LOCATION, AND A TO LOCATION CONTAINS THE OPERAND. PART 1 OF THE PROGRAM SELECTS FROM, TO, AND OPERAND FROM A RANDOM NUMBER GENERATOR, WITH THE OPTION OF HOLDING ANY OR ALL CONSTANT. PART 2 USES A FIXED SET OF FROM, TO, AND OPERAND NUMBERS.

2. REQUIREMENTS

2.1 EQUIPMENT

ONE PDP-8E EQUIPPED WITH TELETYPE.

2.2 STORAGE

THIS PROGRAM USES LOCATIONS 0000-7600(8), THE BINARY LOADER MUST BE STORED IN THE LAST MEMORY PAGE.

2.3 PRELIMINARY PROGRAM

MAINDEC-8E-00A(N), AND MAINDEC-8E-00B(N) MUST HAVE RUN SUCCESSFULLY.

3. LOADING PROCEDURE

THE STANDARD BINARY LOADER IS USED.

4. STARTING PROCEDURE

4.1 SWITCH SETTINGS

SR0(0) = HALT ON ERROR
SR1(1) = ELIMINATE ERROR PRINTOUTS
SR3 = FIXED FROMS (1)
 RANDOM FROMS (0)
SR4 = FIXED TOS (1)
 RANDOM TOS (0)
SR5 = FIXED OPERAND (1)
 RANDOM OPERAND (0)
SR9(0) = DO ONE ISZ ONLY
SR11(1)= DO TEST PART 2 SR3, 4, 5, MUST BE 0'S
SR11(0)= DO TEST PART 1

4.2 STARTING ADDRESS

4,3 OPERATOR ACTION

- A. SET SR (SWITCH REGISTER) TO 0200 AND PRESS LOAD ADDRESS.
- B. SET SR TO DESIRED MODE OF OPERATION; FOR MOST RUNS, SR9=0
ALLOWS THE MOST TESTING IN THE LEAST AMOUNT OF TIME.

FOR FIXED FROM, TO, OR OPERAND USAGE, THE FIXED NUMBER MAY
BE SELECTED AND ENTERED INTO THE MEMORY LOCATIONS SHOWN
BELOW:

FROM	=0002
TO	=0021
OPERAND	=0022

C. PRESS, CLEAR AND THEN CONTINUE.

5. OPERATING PROCEDURE

SAME AS PARAGRAPH 4.

6. ERRORS

6.1 ERROR HALTS AND DESCRIPTION

C(PC)	CAUSE
0002	PERIPHERAL INTERRUPT
0254	HALT ON ERROR, SR0=0

6.2 ERROR PRINTOUTS

F	XXXX	T	YYYY			
0	ZZZZ	F	MMMM	R	NNNN	NS

6.2.1 PRINTOUT EXPLANATION

(FROM)	F XXXX	-THE ISZ INSTRUCTION IN LOCATION XXXX FAILED.
(TO)	T YYYY	-THE OPERAND ADDRESS OF THE ISZ INSTRU- TION WAS YYYY.
(OPERAND)	0 ZZZZ	-THE STARTING COUNT IN THE ISZ LOOP WAS ZZZZ.
(FAILED)	F MMMM	-THE FAILURE OCCURRED TRYING TO ISZ THE NUMBER MMMM.
(RESULT)	R NNNN	-THE RESULT OF THIS ISZ WAS NNNN.
	NS	-NO SKIP OCCURRED
	S,	-INDICATES A SKIP.

6.2.2 EXAMPLES

A. THE FOLLOWING IS A TYPICAL ERROR PRINTOUT.

```
F 3003 T 5470  
0 3705 F 4777 R 5000 S
```

LINE 1 OF THE PRINTOUT IS A STATEMENT OF THE PROBLEM. IT SAYS THAT LOCATED AT 3003 IS AN ISZ INSTRUCTION INCREMENTING AN OPERAND STORED IN LOCATION 5470. LINE 2 OF THE PRINTOUT GIVES INFORMATION FOR ERROR ANALYSIS. 3705 WAS THE INITIAL OPERAND, 4777 WAS THE OPERAND BEING INCREMENTED WHEN THE ERROR OCCURRED, AND 5000 IS THE OPERAND FOLLOWING THE FAILING INCREMENT. THE S INDICATES THAT THE INCREMENT RESULTED IN A SKIP. THE ERROR HERE IS OBVIOUSLY THAT THE SKIP SHOULD NOT HAVE OCCURRED.

B. THE FOLLOWING IS ANOTHER TYPICAL ERROR PRINTOUT.

```
F 3003 T 5470  
0 3705 F 4777 R 5020 NS
```

THIS IS IDENTICAL TO EXAMPLE (A) EXCEPT THAT A DIFFERENT TYPE OF ERROR HAS OCCURRED. THE RESULT OF INCREMENTING 4777 SHOULD BE 5000, NOT 5020.

6.3 ERROR RECOVERY

THE PROGRAM CONTINUES ON, FOLLOWING AN ERROR PRINTOUT UNLESS SR0=0. AFTER A HALT ON ERROR, PUSH CONTINUE TO RESUME TESTING. WHEN ERRORS EXIST, A FAILING CONDITION CHOSEN FROM THOSE TYPED OUT MUST BE USED WITH THE SCOPE MODE. FOR THE SCOPE MODE, PERFORM THE FOLLOWING STEPS:

- A. STOP THE PROGRAM.
- B. INSERT CHOSEN FROM INTO LOCATION 0002.
- C. INSERT CHOSEN TO INTO LOCATION 0021.
- D. INSERT CHOSEN FAILING OPERAND INTO LOCATION 0022
- E. RESTART PROGRAM WITH CONTROL SWITCHES 1,3,4,5, SET TO 1 AND 9 SET TO A 0.

NOTE! BY SETTING SR0 TO A 0, THE PROGRAM HALTS FOLLOWING THE ERROR PRINTOUT. THE OPERATOR MAY AT THIS TIME SET SWITCHES 1, 3, 4, 5, TO A 1 AND 9 TO A 0 AND PUSH CONTINUE. THE PROGRAM ENTERS A SCOPE MODE USING THE FAILING CONDITIONS JUST PRINTED.

7. RESTRICTIONS

7.1 STARTING RESTRICTIONS

NONE.

7.2 OPERATING RESTRICTIONS

THE INTERRUPT IS ENABLED DURING PROGRAM OPERATION. ANY ATTACHED
DEVICE WHICH MIGHT CAUSE SPURIOUS INTERRUPTS, MUST BE DISABLED.

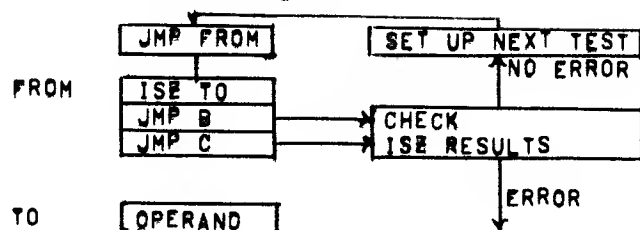
8. MISCELLANEOUS

8.1 EXECUTION TIME

SR9 = 1. 11,000 ISZ OPERATIONS/SECOND.
SR9 = 0. 3,500 ISZ OPERATIONS/SECOND.

PROGRAM DESCRIPTION

THE TEST LOOP IS SHOWN BELOW:



PART 1 OF THE PROGRAM USES A RANDDM NUMBER GENERATOR TO SELECT THE FROM, TO, AND OPERAND NUMBERS. ONCE SELECTED, THE OPERAND IS INCREMENTED UNTIL IT REACHES ZERO. EACH ISZ IS CHECKED BY DUPLICATING ISZ WITH TAD, IAC, DCA. EACH ITERATION IS ALSO CHECKED FOR THE PROPER SKIP OR ND-SKIP CONDITION.

PART 2 OF THE PROGRAM IS ACTUALLY PART 1, WITH THE RANDOM NUMBER GENERATED REPLACED BY A FIXED NUMBER GENERATOR. SEQUENCING OF EVENTS IS AS FOLLOWS:

(NOTE: 621(8)<MEMORY TEST AREA<7600(8)):

- A. FROM = 621 TO = 624 TEST A SET OF 24 SELECTED OPERANDS. TO SAVE TIME IT IS SUGGESTED THAT SR9 = 0, SO THAT THE ISZ IS PERFORMED ON EACH OPERAND ONLY ONCE INSTEAD OF INCREMENTING IT UNTIL THE ISZ INSTRUCTION SKIPS.
- B. FROM = 621 TO = 625 REPEAT THE SET OF OPERANDS USED IN (A) ABOVE.

THIS SEQUENCE CONTINUES UNTIL TO REACHES THE UPPER LIMIT OF THE MEMDRY TEST AREA, FROM IS THEN INCREMENTED BY 1 AND THE PROCESS IS REPEATED. WHEN FROM REACHES THE UPPER LIMIT OF THE MEMORY TEST AREA, THE TEST IS COMPLETE.

IDEALLY, IT IS DESIRABLE TO ISZ EVERY LOCATION FROM EVERY OTHER LOCATION IN THE TEST AREA AND, IN DOING SO, USE ALL 24 OF THE SELECTED WORST CASE OPERANDS FOR EACH SET OF ADDRESSES. THIS IS WHAT PART 2 DOES, BUT IT TAKES MANY DAYS TO COMPLETE THE TEST. IT IS FOR THIS REASON THAT THE PROGRAM USES THE RANDDM NUMBER GENERATOR SYSTEM OF PART 1. PART 2 IS AN ADDITIONAL FEATURE OF THE PROGRAM WITH VERY LIMITED USE.

A FC IS PRINTED AFTER EACH GROUP OF 32,000 TESTS.

/POP-8E ISZ TEST

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/

/CONSTANTS AND VARIABLES

*0

0000	0000	0	
0001	5001	JMP 1	/PERIPHERAL INTERRUPT
0002	0002	FRMLOC, 2	/ISZ TEST INSTRUCTION LOCATION
0003	0003	LIMLO, 3	/LOW LIMIT TEST AREA
0004	0000	0	
0005	0000	0	
0006	0202	LIMHI, -7576	/HIGH LIMIT TEST AREA
0007	0547	ASUC, SUC	
0010	0007	MSK7, 0007	/OCTAL CONVERSION MASK
0011	0000	WORK, 0	/IR0
0012	0000	WORK1, 0	/IR1
0013	7401	M377, -377	
0014	3607	NUM, 3607	/THE RANDOM NUMBER LOCATION
0015	0003	THREE, 3	
0016	2421	ISZ1, ISZ I TOLOC	/MOVING ISZ
0017	5116	JMP1, JMP BACK	/TEST INSTRUCTION
0020	5141	JMP2, JMP BAKBRN	/GROUP.
0021	0000	TOLOC, 0	/LOCATION TO BE ISZ'0
0022	0000	PATRN, 0	/STARTING ISZ PATTERN
0023	0000	BEFOR, 0	/FAILING PATTERN BEFORE FAILING ISZ
0024	0000	AFTER, 0	/PREDICTED RESULTS OF EACH ISZ
0025	0004	K4, 4	/SWITCH REGISTER MASKS
0026	0400	K0400, 0400	
0027	0200	K0200, 0200	
0030	0100	K0100, 0100	
0031	0000	NOTE, 0	/7'S=ERROR WITH NO SKIP
0032	0257	PRINT, INF1-1	/0'S=ERROR WITH SKIP
0033	0201	AERR1, ERR1	
0034	0206	AERR2, ERR2	
0035	0413	APOR, PDR	
0036	1014	ITAONM, TAD NUM	
0037	0600	ATFCLF, TFCLF	

/SR0(0)=HALT AFTER ERROR PRINTOUT

/SR1(1)=NO PRINTOUTS

/SR3(1) = HOLD FROM CONSTANT

/SR4(1) = HOLD TO CONSTANT

/SR5(1) = HOLD PATTERN CONSTANT

/SR9(0) = 00 ONE ISZ ONLY

/SR11(1) = 00 TEST PART 2

/

/

/PROGRAM START

0040	4441	START, JMS I .+1	/ION
0041	0614	PATCH	/LAS
0042	0015	AND THREE	

0043	7640	SEA CLA	/SKIP IF PART 1
0044	5426	JMP I K0400	/GO TO PART 2
0045	1036	TAD ITADNM	
0046	3165	DCA RANUM+1	
		/CHECK FOR FIXED PATTERN	
0047	7604	CHEK1, LAS	
0050	0030	AND K0100	
0051	7440	SEA	
0052	5055	JMP CHEK2	

0053	4164	SELPAT, /SELECT THE PATTERN	
0054	3022	JMS RANUM	
		DCA PATRN	

0055	7604	CHEK2, /CHECK FOR FIXED TO	
0056	0027	LAS	
0057	7640	AND K0200	
0060	5065	SEA CLA	
		JMP CHEK3	

0061	4164	SELTO, /SELECT THE TO LOCATION	
0062	3021	JMS RANUM	
0063	1021	DCA TOLOC	
0064	4151	TAD TOLOC	
		JMS LINTST	

0065	7604	CHEK3, /CHECK FOR FIXED FROM	
0066	0026	LAS	
0067	7640	AND K0400	
0070	5075	SEA CLA	
		JMP PLCINT	

0071	4164	SELFRM, /SELECT THE FROM LOCATION	
0072	3002	JMS RANUM	
0073	1002	DCA FRMLOC	
0074	4151	TAD FRMLOC	
		JMS LINTST	

/PLACE FROM INSTRUCTIONS

0075	7240	PLCINT, CLA CMA	
0076	1002	TAD FRMLOC	
0077	3011	DCA WORK	
0100	1010	TAD I021	
0101	3411	DCA I WORK	
0102	1017	TAD JMP1	
0103	3411	DCA I WORK	
0104	1020	TAD JMP2	
0105	3411	DCA I WORK	

0106	1022	/DEPOSIT PATTERN IN TO LOCATION	
0107	3421	TAD PATRN	
		DCA I TOLDC	

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                                /STORE PREDICTED ISZ RESULT
0110 1022                      TAD PATRN
0111 3023                      DCA BEFOR
0112 1023    LUP1,            TAD BEFOR
0113 7001                      IAC
0114 3024                      DCA AFTER
0115 5407                      JMP I ASUC

                                /RETURN FOR NO SKIP CONDITION
0116 7604    BACK,          LAS
0117 7004                      RAL
0120 7710                      SPA CLA
0121 5132                      JMP LAS1
0122 1421                      TAD I TOLOC
0123 7041                      CIA
0124 1024                      TAD AFTER
0125 7640                      SZA CLA
0126 5433                      JMP I AERR1    /ERROR IN ISZ OPERATION
0127 1421                      TAD I TOLOC
0130 7650                      SNA CLA
0131 5433                      JMP I AERR1    /ERROR IN ISZ SKIP DETECTION
0132 7604    LAS1,          LAS
0133 0025                      AND K4
0134 7650                      SNA CLA    /SKIP IF NOT ONE ISZ (SR9)
0135 5047                      JMP CHEK1
0136 7001                      IAC
0137 1023                      TAD BEFOR
0140 5111                      JMP LUP1-1

                                /RETURN FOR SKIP CONDITION
0141 7604    BAKBRN,        LAS
0142 7004                      RAL
0143 7710                      SPA CLA
0144 5047                      JMP CHEK1
0145 1421                      TAD I TOLOC
0146 7640                      SZA CLA    /SKIP IF TO LOCATION OK
0147 5434                      JMP I AERR2    /ERROR IN ISZ LOCATION
0150 5047                      JMP CHEK1

                                /TEST HIGH-LOW LIMITS
0151 0000    LIMITST, 0
0152 7510                      SPA
0153 5160                      JMP ,+5
0154 1003                      TAD LIMLO
0155 7700                      SMA CLA
0156 5551                      JMP I LIMITST
0157 5165                      JMP RANUM+1
0160 1006                      TAD LIMHI
0161 7700                      SMA CLA
0162 5165                      JMP RANUM+1
0163 5551                      JMP I LIMITST

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0164 0000      RANUM, 0      /RANDOM NUMBER GENERATOR
0165 1014      TAD NUM
0166 7104      RAL CLL
0167 7430      SEL
0170 1015      TAD THREE
0171 3014      DCA NUM
0172 1014      TAD NUM      /AC=NEW RANDOM NUMBER
0173 5564      JMP I RANUM

0174 1000      K1000, 1000
0175 0000      KP, 0

      0200      *200
0200 5040      JMP      START
                        /ERROR ROUTINE 1
0201 1340      ERR1, TAD SKPDAT+6
0202 3332      DCA SKPDAT
0203 7040      CMA
0204 3031      DCA NOTE
0205 5210      JMP KPGO

                        /ERROR ROUTINE 2
0206 1331      ERR2, TAD SKPDAT-1
0207 3332      DCA SKPDAT
0210 1002      KPGO, TAD FRMLOC
0211 3011      DCA WORK
0212 1370      TAD A3
0213 4342      JMS SETUP

0214 1021      TAD TOLOC
0215 3011      DCA WORK
0216 1371      TAD A4
0217 4342      JMS SETUP

0220 1022      TAD PATRN
0221 3011      DCA WORK
0222 1372      TAD A5
0223 4342      JMS SETUP
0224 1023      TAD BEFOR
0225 3011      DCA WORK
0226 1373      TAD A6
0227 4342      JMS SETUP

0230 1421      TAD I TOLOC
0231 3011      DCA WORK
0232 1374      TAD A7
0233 4342      JMS SETUP

      /TTY PRINT ROUTINE
0234 6002      TTY, IOF
0235 1032      TAD PRINT
0236 3011      DCA WORK
0237 1411      TAD I WORK
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0240 6046      TLS
0241 6041      TSF
0242 5241      JMP .-1
0243 1013      TAD M377
0244 7640      SZA CLA
0245 5237      JMP TTY+3
0246 6042      TCF
0247 6001      ION
0250 7604      LAS
0251 7700      SMA CLA
0252 7402      HLT                /HALT AFTER ERROR (SR0)

0253 1031      TAD NOTE
0254 7650      SNA CLA
0255 5047      JMP CHEK1
0256 3031      DCA NOTE
0257 5132      JMP LAS1                /RETURN TO NO SKIP ROUTINE

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0260 0306      /ERROR PRINT OUT LINE 1
0261 0240      INF1, 306          /F FROM (INSTRUCTION LOCATION)
0262 0000      240              /SPACE
0263 0000      INDATA, 0        /X LOCATION
0264 0000      0               /X
0265 0000      0               /X
0266 0240      0               /X
0267 0240      240             /SPACE
0270 0324      240             /SPACE
0271 0240      324             /T TO (OPERAND ADDRESS)
0272 0000      240             /SPACE
0273 0000      ONDATA, 0        /X ADDRESS
0274 0000      0               /X
0275 0000      0               /X
0276 0215      0               /X
0277 0212      215            /CR
0300 0215      212            /LF
0301 0215      215            /CR
0301 0215      215            /CR

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0302 0317      /ERROR PRINTOUT LINE 2
0303 0240      317            /O OPERAND (STARTING COUNT)
0304 0000      240            /SPACE
0305 0000      STDATA, 0       /X PATTERN
0306 0000      0              /X
0307 0000      0              /X
0310 0240      0              /X
0311 0240      240            /SPACE
0312 0306      240            /SPACE
0313 0240      306            /F FAILING COUNT
0314 0000      240            /SPACE
0315 0000      FLDATA, 0       /X PATTERN BEFORE FAILING ISZ
0316 0000      0              /X
0317 0000      0              /X
0320 0240      0              /X
0320 0240      240            /SPACE

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0321	0240	240	/SPACE	
0322	0322	322	/R	RESULT AFTER FAILURE
0323	0240	240	/SPACE	

0324	0000	RSDATA, 0	/X	PATTERN AFTER FAILING ISZ
0325	0000	0	/X	
0326	0000	0	/X	
0327	0000	0	/X	
0330	0240	240	/SPACE	
0331	0240	240	/SPACE	
0332	0316	SKPDAT, 316	/N	NO
0333	0323	323	/S	SKIP
0334	0215	215	/CR	
0335	0212	212	/LF	
0336	0212	212	/LF	
0337	0377	377	/RUBOUT	
0340	0316	316	/N	
0341	0323	323	/S	

0342	0000	SETUP, 0	
0343	3012	DCA WORK1	
0344	1011	TAD WORK	
0345	7006	RTL	
0346	7006	RTL	
0347	4362	JMS MORSU	
0350	7012	RTR	
0351	7012	RTR	
0352	7012	RTR	
0353	4362	JMS MORSU	
0354	7012	RTR	
0355	7010	RAR	
0356	4362	JMS MORSU	
0357	4362	JMS MORSU	
0360	7200	CLA	
0361	5742	JMP I SETUP	
0362	0000	MORSU, 0	
0363	0010	AND MSK7	
0364	1375	TAD TW6	
0365	3412	DCA I WORK1	
0366	1011	TAD WORK	
0367	5762	JMP I MORSU	

0370	0261	A3,	/PAGE 1 CONSTANTS
0371	0271	A4,	INDATA-1
0372	0303	A5,	ONDATA-1
0373	0313	A6,	STDATA-1
0374	0323	A7,	FLDATA-1
0375	0260	TW6,	RSDATA-1
			0260

	0400	/PART 2 INITIALIZATION ROUTINE	
0400	1003	*400	
			TAD LIMLO

0401	7041		CIA	
0402	3310		DCA FROM	/LOW LIMIT TO FROM
0403	1003		TAD LIMLO	
0404	7040		CMA	
0405	3311		DCA TO	
0406	1346		TAD A0	
0407	3313		DCA PATCYC	
0410	1314		TAD INST1	
0411	3165		DCA RANUM+1	
0412	5047		JMP CHEK1	/GO TO PAGE 0 START
0413	1164	PDR,	/PATH DECISION ROUTINE	
0414	7041		TAD RANUM	
0415	1305		CIA	
0416	7650		TAD GFROM	
0417	5303		SNA CLA	/SKIP IF NOT REQUESTING FROM
			JMP FRUT	/GO TO FROM ADDRESS ROUTINE
0420	1164		TAD RANUM	
0421	7041		CIA	
0422	1306		TAD GTO	
0423	7650		SNA CLA	/SKIP IF NOT REQUESTING TO
0424	5301		JMP TORUT	/GO TO TO ADDRESS ROUTINE
0425	5226		JMP PRUT	/GO TO PATTERN ROUTINE
0426	1713	PRUT,	/SELECT PATTERN AND OTHER THINGS	
0427	3312		TAD I PATCYC	
0430	1312		DCA PATT	
0431	7450		TAD PATT	
0432	5240		SNA	/NO SKIP IF END OF PATTERN TABLE
0433	7201		JMP .+6	/END PATTERN TABLE LOOK AROUND
0434	1313		CLA IAC	
0435	3313		TAD PATCYC	
0436	1312		DCA PATCYC	
0437	5564		TAD PATT	
			JMP I RANUM	/RETURN, AC=NEW PATTERN
			/	
0440	1345		TAD AK7776	
0441	3313		DCA PATCYC	/RESTOR START ADDRESS OF PATT. TABLE
0442	7001		IAC	
0443	1311		TAD TO	
0444	3311		DCA TO	/INCREMENT TO
0445	1311		TAD TO	
0446	7041		CIA	
0447	1310		TAD FROM	
0450	7640		SZA CLA	/SKIP IF TO = FROM
0451	5255		JMP .+4	
0452	1311		TAD TO	
0453	1015		TAD THREE	
0454	3311		DCA TO	/SKIP AROUND FROM
0455	1311		TAD TO	
0456	7500		SMA	
0457	5276		JMP GDUT	

0460	1006		TAD LIMHI	
0461	7710		SPA CLA	/SKIP IF END TEST AREA
0462	5276		JMP GOUT	
0463	7201		CLA IAC	
0464	1310		TAD FROM	
0465	3310		DCA FROM	/ADVANCE FROM
0466	1003		TAD LIMLO	
0467	7041		CIA	
0470	3311		DCA TO	/RESET TO ADDRESS
0471	1310		TAD FROM	
0472	1006		TAD LIMHI	
0473	7710		SPA CLA	
0474	5276		JMP GOUT	
0475	5200		JMP 400	
0476	7200	GOUT,	CLA	
0477	1312		TAD PATT	
0500	5564		JMP I RANUM	
0501	1311	TORUT,	/SELECT TO ROUTINE	
0502	5564		TAD TO	
			JMP I RANUM	
0503	1310	FRUT,	/SELECT FROM ROUTINE	
0504	5564		TAD FROM	
			JMP I RANUM	
0505	0072	GFROM,	SELFRM+1	/PAGE 3 CONSTANTS
0506	0062	GTO,	SELTO+1	/STORED RETURN ADDRESS WHEN
0507	0054	GPAT,	SELPAT+1	/RANDOM FROM IS REQUESTED
				/STORED RETURN ADDRESS WHEN
				/RANDOM TO IS REQUESTED
				/STORED RETURN ADDRESS WHEN
				/RANDOM PATTERN IS REQUESTED
0510	0000	FROM,	0	/CURRENT FROM ADDRESS
0511	0000	TO,	0	/CURRENT TO ADDRESS
0512	0000	PATT,	0	/CURRENT PATTERN
0513	0000	PATCYC,	0	/CURRENT PATTERN ADDRESS
0514	5435	INST1,	JMP I APDR	
0515	7776	K7776,	7776	
0516	7775		7775	
0517	7773		7773	
0520	7767		7767	
0521	7757		7757	
0522	7737		7737	
0523	7677		7677	
0524	7577		7577	
0525	7377		7377	
0526	6777		6777	
0527	5777		5777	
0530	3777		3777	
0531	0001		0001	
0532	0003		0003	
0533	0007		0007	
0534	0017		0017	

0535	0037		0037
0536	0077		0077
0537	0177		0177
0540	0377		0377
0541	0777		0777
0542	1777		1777
0543	3777	K3777,	3777
0544	0000		0
0545	0515	AK7776,	K7776
0546	0544	A0,	K3777+1

0547	1375	SUC,	TAD CT
0550	7001		IAC
0551	3375		DCA CT
0552	1375		TAD CT
0553	7640		SZA CLA
0554	5437		JMP I ATFCLF
0555	1175		TAD KP
0556	1174		TAD K1000
0557	3175		DCA KP
0560	1175		TAD KP
0561	7640		SZA CLA
0562	5437		JMP I ATFCLF
0563	6002		IDF
0564	1376		TAD INF2
0565	3011		DCA WORK
0566	5767		JMP I .+1
0567	7602		7602
0570	0215		215
0571	0212		212
0572	0306		306
0573	0303		303
0574	0377		377
0575	0000	CT,	0
0576	0567	INF2,	567

0600 *600

/CHECK FOR TO=FROM CONFLICT

0600	1021	TFCLF,	TAD TOLOC
0601	7041		CIA
0602	1002		TAD FRMLC
0603	7450		SNA
0604	5055		JMP CHEK2
0605	7001		IAC
0606	7450		SNA
0607	5055		JMP CHEK2
0610	7001		IAC
0611	7650		SNA CLA
0612	5055		JMP CHEK2

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0613 5402          JMP I FRMLOC

0614 0000          PATCH, 0          /RESTORE THEN GO AWAY
0615 3000          DCA 0
0616 1232          TAD X
0617 3001          DCA 1
0620 1233          TAD X1
0621 3002          DCA 2
0622 1234          TAD X2
0623 3003          DCA 3
0624 1235          TAD X3
0625 3040          DCA START
0626 1236          TAD X4
0627 3041          DCA START+1
0630 6001          ION
0631 5614          JMP I PATCH

0632 7402          X,      7402
0633 0000          X1,    0
0634 7157          X2,    7157
0635 6001          X3,    ION
0636 7604          X4,    LAS

          7602      *7602
7602 1411          TAD I WORK
7603 6046          TLS
7604 6041          TSF
7605 5204          JMP , -1
7606 1013          TAD M377
7607 7640          SZA CLA
7610 5202          JMP , -6
7611 5217          JMP OVR

          7617      *7617
7617 6042          OVR,    TCF
7620 6001          ION
7621 5437          JMP I ATFCLF

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S